11

Chapter

Numerical Solution of Partial Differential Equations

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11.1 INTRODUCTION

Partial differential equations mostly occur in various branches of applied mathematics; *e.g.*, hydrodynamics magneto-hydrodynamics; elasticity, quantum-mechanics electromagnetic theory, etc. The solutions to these equations are obtained by standard methods as given in various texts of differential equations. On the other hand, it is generally easier to obtain sufficiently approximate solutions by simple and efficient numerical methods. Many numerical methods have been proposed to obtain sufficiently approximate solutions to partial differential equations. But, only the methods of finite differences have become most popular and hence are used more gainfully than others. We will therefore study such methods involving finite differences to obtain approximate solutions, numerically, of various simple problems occurring in partial differential equations. A general second-order linear partial differential equation is of the type.

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} + D\frac{\partial u}{\partial x} + E\frac{\partial u}{\partial y} + Fu = G. \qquad \text{... (1)}$$

where A, B, C, D, E, F, G are all functions of x and y. This equation can be classified with respect to the sign of the discriminant.

$$\Delta s = B^2 - 4AC.$$

(*i*) If $\Delta s < 0$, at a point (x, y) in the plane, the equation is said to be of elliptic-type.

(*ii*) If $\Delta s > 0$, at a point (x, y) in the plane, the equation is said to be of hyperbolic-type.

(*iii*) If $\Delta s = 0$, at a point (x, y) in the plane, the equation is said to be of parabolic-type.

Consider the following partial differential equations

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \textbf{(Laplace's equation)} \qquad \text{... (i)}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} \quad \text{(Wave equation)} \qquad \dots (ii)$$

and

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t} \quad \text{(Heat conduction equation)} \qquad \dots (iii)$$

where (x, y) are the space co-ordinates and t is the time co-ordinate. In equation (i) we have $A = 1$, $B = 0$, $C = 1$;

$\Rightarrow$ $\Delta s = B^2 - 4AC = -4$ = Negative. Hence Laplace's equation is of elliptic type.

In equation (ii), we also have Δs = positive, *i.e.*, $\Delta s > 0$ and hence the equation is hyperbolic type. On the other hand, in equation (iii) we have $\Delta s = 0$ and hence the heat equation is parabolic type. While studying partial differential equations, these are three types of problems.

Type 1. Dirichlet's Problem.

Let there exist a continuous function f on the boundary C of a region R. To find the function u satisfying Laplace's equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ in R and $u = f$ on C.

Type 2. Cauchy's Problem.

To find the function u satisfying the equation $\frac{\partial^2 u}{\partial t^2} + \frac{\partial^2 u}{\partial x^2} = 0$; $t > 0$. $u(x, 0)$. $f(x)$ and $[\partial u / \partial t]_{x,0} = g(x)$ where $f(x)$ and $g(x)$ are arbitrary.

Type 3. $\left(\frac{\partial u}{\partial t}\right) - \left(\frac{\partial^2 u}{\partial x^2}\right) = 0, t > 0$

$u(x, 0) = f(x)$. To find $u(x, t)$

These partial differential equations possess unique solutions.

SOLVED EXAMPLES

Example 11.1. **Classify the following partial differential equations:**

(a) $2\frac{\partial^2 u}{\partial x^2} + 4\frac{\partial^2 u}{\partial x \partial y} + 3\frac{\partial^2 u}{\partial y^2} = 2$ (b) $\frac{\partial^2 u}{\partial x^2} + 4\frac{\partial^2 u}{\partial x \partial y} + 4\frac{\partial^2 u}{\partial y^2} = 0$

(c) $5\frac{\partial^2 u}{\partial x^2} - 9\frac{\partial^2 u}{\partial x \partial t} + 4\frac{\partial^2 u}{\partial t^2} = 0.$

Solution: (a) Comparing this equation with (1)

$$A = 2, B = 4, C = 3$$

$$\therefore \quad B^2 - 4AC = (4)^2 - 4 \times 2 \times 3 = -8 < 0$$

showing that the given equation is elliptic at all points.

(b) Comparing this equation with (1)

$$\therefore \quad A = 1, B = 4, C = 4$$
$$B^2 - 4AC = 4^2 - 4 \times 4 = 0$$

showing that the given equation is parabolic at all points.

(c) Comparing this equation with (1)

$$\therefore \quad A = 5, B = -9, C = 4$$
$$B^2 - 4AC = (-9)^2 - 4 \times 5 \times 4$$
$$= 1 > 0$$

showing that the given equation is hyperbolic of all points.

Example 11.2. **Find whether the following operators are hyperbolic, parabolic and elliptic.**

(a) $x^2 \dfrac{\partial^2 u}{\partial t^2} - \dfrac{\partial^2 u}{\partial x^2} + u$ (b) $t\dfrac{\partial^2 u}{\partial t^2} + 2\dfrac{\partial^2 u}{\partial x \partial t} + x\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial u}{\partial x}$ (c) $x\dfrac{\partial^2 u}{\partial x^2} + t\dfrac{\partial^2 u}{\partial x \partial t} + \dfrac{\partial^2 u}{\partial t^2}$

Solution: (a) Here $\quad A = x^2, B = 0, C = -1$

Now $\quad B^2 - 4AC = 0^2 - 4x^2 . (-1) = 4x^2$

$\therefore$ The operator is hyperbolic if $4x^2 > 0$, *i.e.*, $x > 0$

The operator is parabolic if $4x^2 = 0$, *i.e.*, $x = 0$

Since $4x^2$ cannot be negative (being a square), hence operator cannot be elliptic.

(b) Here $\quad A = 1, B = 2, C = x$

Now $\quad B^2 - 4AC = 4 - 4tx$

$\therefore$ The operator is hyperbolic if $4 - 4tx > 0$, *i.e.*, $tx < 1$

The operator is elliptic if $4 - 4tx < 0$, *i.e.*, $tx > 1$

The operator is parabolic if $4 - 4tx = 0$, *i.e.*, $tx = 1$.

(c) Here $\quad A = x, B = t, C = 1$

Now $\quad B^2 - 4AC = t^2 - 4 \times x \times 1 = t^2 - 4x$

$\therefore$ The operator is hyperbolic if $t^2 - 4x > 0$, *i.e.*, $t^2 > 4x$

The operator is elliptic if $t^2 - 4x < 0$, *i.e.*, $t^2 < 4x$

The operator is parabolic if $t^2 - 4x = 0$, *i.e.*, $t^2 = 4x$

Example 11.3. **Classify the equation**

$$(1 - x^2)\frac{\partial^2 z}{\partial x^2} - 2xy\frac{\partial^2 z}{\partial x \partial y} + (1 - y^2)\frac{\partial^2 z}{\partial y^2} + x\frac{\partial z}{\partial x} + 3x^2 y\frac{\partial z}{\partial y} - 2z = 0.$$

Solution: Here $\quad A = 1 - x^2, B = -2xy, C = 1 - y^2$

$$\therefore \quad B^2 - 4AC = (-2xy)^2 - 4(1 - x^2)(1 - y^2)$$
$$= 4x^2y^2 - 4(1 - x^2)(1 - y^2)$$
$$= 4(-1 + x^2 + y^2)$$

The equation is elliptic if $B^2 - 4AC < 0$, *i.e.*, $x^2 + y^2 < 1$

The equation is parabolic if $B^2 - 4AC = 0$, *i.e.*, $x^2 + y^2 = 1$

The equation is hyperbolic if $B^2 - 4AC > 0$, *i.e.*, $x^2 + y^2 > 1$

Example 11.4. Show that the equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ is hyperbolic.

Solution: The given equation is

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0$$

Here $\quad A = 1., B = 0, C = -c^2$

Now, $\quad B^2 - 4AC = (0)^2 - 4(1)(-c^2) = 4c^2 > 0$

Hence, $\quad B^2 - 4AC > 0$

$\therefore$ The given equation is hyperbolic.

EXERCISE 11.1

Classify the following operators:

1. $\frac{\partial^2 u}{\partial t^2} + 4\frac{\partial^2 u}{\partial x \partial t} + 4\frac{\partial^2 u}{\partial x^2}$ [Ans. Parabolic]

2. $\frac{\partial^2 u}{\partial t^2} + \frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 u}{\partial x^2}$ [Ans. Elliptic]

Classify the following:

3. $t\frac{\partial^2 u}{\partial t^2} + 3\frac{\partial^2 u}{\partial x \partial y} + x\frac{\partial^2 u}{\partial x^2} + 17\frac{\partial u}{\partial x}$.

$$\left[\begin{array}{ll} \textbf{Ans.} & \text{Hyperbolic if } xt < \frac{9}{4} \\ & \text{Parabolic if } xt = \frac{9}{4} \\ \text{and} & \text{Elliptic if } xt > \frac{9}{4} \end{array}\right]$$

4. $\frac{\partial^2 u}{\partial t^2} + t\frac{\partial^2 u}{\partial x \partial t} + x\frac{\partial^2 u}{\partial x^2}$

$$\left[\begin{array}{ll} \textbf{Ans.} & \text{Hyperbolic if } t^2 > 4x \\ & \text{Parabolic if } t^2 = 4x \\ \text{and} & \text{Elliptic if } t^2 < 4x \end{array}\right]$$

5. Classify the equation

$$x^2\frac{\partial^2 u}{\partial t^2} + 3\frac{\partial^2 u}{\partial x \partial t} + x\frac{\partial^2 u}{\partial x^2} + 17\frac{\partial u}{\partial t} = 100u.$$

$$\left[\begin{array}{ll} \textbf{Ans.} & \text{Hyperbolic if } 9 - 4x^3 > 0 \\ & \text{Parabolic if } 9 - 4x^3 = 0 \\ \text{and} & \text{Elliptic if } 9 - 4x^3 < 0 \end{array}\right]$$

11.2 FINITE-DIFFERENCE APPROXIMATIONS OF PARTIAL DERIVATIVES

Let the $x-y$ plane be divided into a network of rectangles of side $\Delta_x = h$ and $\Delta_y = k$ by drawing the set of lines $x = ih$ ($i = 0, 1, 2, ...$) and $y = jk$ ($j = 0, 1, 2, ...$). The points of intersection of these lines are called mesh points (lattice points or gird-points). Now proceeding as in Boundary-value problems one may obtain

$$\frac{\partial u}{\partial x} = \frac{u_{i+1,j} - u_{i,j}}{h} + 0(h) \quad \text{... (i)}$$

$$= \left[\frac{u_{i,j} - u_{i-1,j}}{h}\right] + 0(h) \quad \text{... (ii)}$$

$$= \frac{u_{i+1,j} - u_{i-1,j}}{2h} + 0(h^2) \quad \text{... (iii)}$$

and
$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial^2 u}{\partial x^2}$$

$$= \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2} + 0(h^2) \quad \text{... (iv)}$$

where
$$u_{i,j} = u_{(ih,\, jk)}\, u_{(x,\, y)}$$

similarly, we have

$$\frac{\partial u}{\partial y} = \frac{u_{i,j+1} - u_{i,j}}{k} + 0(k) \quad \text{... (v)}$$

$$= \frac{u_{i,j} - u_{i,j-1}}{k} + 0(k) \quad \text{... (vi)}$$

$$= \frac{u_{i,j+1} - u_{i,j-1}}{2k} + 0(k^2) \quad \text{... (vii)}$$

and
$$\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial^2 u}{\partial y^2}$$

$$= \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{k^2} + 0(k^2) \quad \text{... (viii)}$$

To obtain finite-difference analogues of partial differential equations.

Laplace's Equation
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{... (ix)}$$

Putting the values of $\frac{\partial^2 u}{\partial x^2}$ and $\frac{\partial^2 u}{\partial y^2}$ from (*iv*) and (*viii*) in (*ix*); we get

$$\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2} + \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{k^2} = 0 \quad \text{... (x)}$$

If $h = k$, we have

$$u_{i,j} = \frac{1}{4}\left[u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}\right] \quad \text{... } (xi)$$

This shows that the value of u at any point is the mean of its values at the four neighbouring points. The above formula is known as the **standard five-point formula and is written sometimes as:**

$$u_{i+1,j} - u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j} = 0 \quad \text{... } (xii)$$

Again, expanding the terms on the right side of (xi) by Taylor's series method, we get

$$u_{i+1,j} - u_{i-1,j+1} + u_{i,j-1} - 4u_{i,j} = h^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) - \frac{1}{6} h^4 \frac{\partial^4 u}{\partial x^2 \partial y^2} + 0(h^6) \quad \text{... } (xiii)$$

Instead of (xi), one may also use the formula

$$u_{i,j} = \frac{1}{4}\left[u_{i-1,j-1} + u_{i+1,j-1} + u_{i+1,j+1} + u_{i-1,j+1}\right] \quad \text{... } (xiv)$$

This formula uses the functional values at the *diagonal-points* and hence is known as the *diagonal-point formula*. We know that the Laplace equation is invariant when the axes are turned through an angle of 45° and hence (xiv) is perfectly valid. Further expanding the R.H.S. of (xiv) by Taylor's series method, we obtain

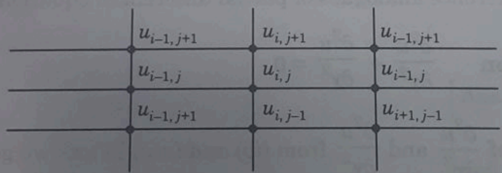
$$u_{i-1,j-1} + u_{i+1,j-1} + u_{i+1,j+1} + u_{i-1,j+1} - 4u_{i,j} = \frac{2}{3} h^4 \frac{\partial^4 u}{\partial x^2 \partial y^2} + 0(h^6) \quad \text{... } (xv)$$

If we neglect the terms of the order h^6, then from $(xiii)$ and (xv) it is obvious that the error in the diagonal formula is four times as compared to the standard formula. It means *that in all computations, we should prefer standard five-point formula.*

Eliminating the term containing h^4 from $(xiii)$ and (xv), we obtain **Nine-point Formula** as:

$$u_{i-1,j-1} + u_{i+1,j-1} + u_{i+1,j+1} + u_{i-1,j+1} - 4(u_{i+1,j} + u_{i-1,j} + u_{i,j-1}) - 20u_{i,j} = 0 \quad \text{... } (xvi)$$

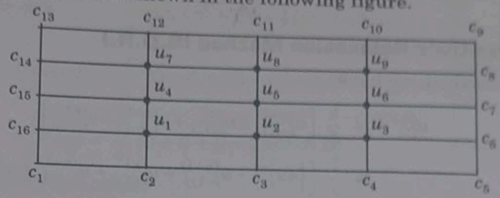
This formula consists of the error of order h^6. On the same lines, we can obtain the finite-difference analogues of the wave equation $\left(\frac{\partial^2 u}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = 0\right)$ and the heat conduction equation $\left(\frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial t} = 0\right)$.



Here our general procedure is to approximate the partial differential equation by a finite-ifference analogue and then to find the solution at the mesh points.

11.3 TO SOLVE LAPLACE'S EQUATION $(\partial^2 u/\partial x^2)+(\partial^2 u/\partial y^2)=0$ IN THE BOUNDED REGION R WITH BOUNDARY C.

Just like Dirichlet's problem, let u be specified everywhere on C. For the sake of convenience, let R denote the square region so that it can be divided into a network of small squares of side h. Further, let the values of $u(x, y)$ on the boundary C be C_1 and further let the interior mesh points and boundary points be as shown in the following figure.



Using diagonal five-point formula: Then Laplace's equation $\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}=0$ can be either replaced by standard five-point formula or by the diagonal five points formula, the approximate function values a the interior points can be easily computed. We thus compute u_5, u_7, u_9, u_1 and u_3 in this order as follows:

$$u_5=\frac{1}{4}(c_1+c_5+c_9+c_{13});\qquad u_7=\frac{1}{4}(c_{15}+u_5+c_{11}+c_{13})$$

$$u_9=\frac{1}{4}(u_5+c_7+c_9+c_{11});\qquad u_1=\frac{1}{4}(c_1+c_3+u_5+c_{15})$$

$$u_3=\frac{1}{4}(c_3+c_5+c_7+u_5)$$

Then we proceed to compute in order, the remaining quantities; u_8, u_4, u_6, u_2 by the standard point formula (11) and thus we get

$$u_8=\frac{1}{4}(u_5+u_9+c_{11}+c_7);\qquad u_4=\frac{1}{4}(u_1+u_5+u_7+c_{15})$$

$$u_6=\frac{1}{4}(u_3+c_7+u_9+u_5);\qquad u_2=\frac{1}{4}(c_3+u_3+u_5+u_1)$$

when all u_i s $(i = 1, 2, 3, \ldots 9)$ are calculated. Their accuracy is improved by any of the following iterative methods.

11.3.1 Point-Jacobi Iterative Method

If $u_{i,j}^{(n)}$ denotes the nth iterative value of $u_{i,j}$ then an iterative procedure to solve.

$$u_{i+1,j}+u_{i-1,j}+u_{i,j+1}+u_{i,j-1}-4u_{i,j}=0 \text{ given by}$$

$$u_{i,j}^{(n+1)}=\frac{1}{4}\left[u_{i-1,j}^{(n)}+u_{i+1,j}^{(n)}+u_{i,j-1}^{(n)}+u_{i,j+1}^{(n)}\right]$$

for the interior mesh point and is known as point Jacobi's method.

11.3.2 Gauss-Seidel Method

The iterative formula is

$$u_{i,j}^{(n+1)} = \frac{1}{4}\left[u_{i-1,j}^{(n)} + u_{i+1,j}^{(n)} + u_{i,j-1}^{(n)} + u_{i,j+1}^{(n)}\right]$$

and this method uses the latest iterative values available. It scans the mesh points systematically from left to right along successive rows. This method converges twice as fast as Jacobi's scheme.

11.3.3 Successive-Over-Relaxation Method (S.O.R.)

By Gauss-Seidel method, we have

$$\begin{aligned}
u_{i,j}^{(n+1)} &= \frac{1}{4}\left[u_{i-1,j}^{(n+1)} + u_{i+1,j}^{(n)} + u_{i,j-1}^{(n+1)} + u_{i,j+1}^{(n)}\right] \\
&= \frac{1}{4}\left[\left\{4u_{i,j}^{(n)} + u_{i-1,j}^{(n+1)} + u_{i+1,j}^{(n)} + u_{i,j-1}^{(n+1)} + u_{i,j+1}^{(n)}\right\} - 4u_{i,j}^{(n)}\right] \\
&= u_{i,j}^{(n)} + \frac{1}{4}\left[u_{i-1,j}^{(n+1)} + u_{i+1,j}^{(n)} + u_{i,j-1}^{(n+1)} + u_{i,j+1}^{(n)} - 4u_{i,j}^{(n)}\right] \\
&= u_{i,j}^{(n)} + \frac{1}{4} R_{i,j} \qquad \text{... (i)}
\end{aligned}$$

where

$$= R_{i,j} = \left[u_{i-1,j}^{(n+1)} + u_{i+1,j}^{(n)} + u_{i,j-1}^{(n+1)} + u_{i,j+1}^{(n)} - 4u_{i,j}^{(n)}\right] \qquad \text{... (ii)}$$

This clearly shows that $R_{i,j}$ is nothing but the change in the value of $u_{i,j}$ for **Gauss-Seidel iteration.** In **S.O.R.** method a larger change then the above is given to $u_{i,j}^{(n)}$ and the corresponding iteration formula becomes:

$$\begin{aligned}
u_{i,j}^{(n+1)} &= u_{i,j}^{(n)} + \frac{1}{4}\,\omega R_{i,j} \\
&= u_{i,j}^{(n)} + \frac{1}{4}\,\omega\left[u_{i-1,j}^{(n+1)} + u_{i+1,j}^{(n)} + u_{i,j-1}^{(n+1)} + u_{i,j+1}^{(n)} - 4u_{i,j}\right] \\
&= u_{i,j}^{(n)}\,.\,(1-\omega) + \frac{\omega}{4}\left[u_{i-1,j}^{(n+1)} + u_{i+1,j}^{(n)} + u_{i,j-1}^{(n+1)} + u_{i,j+1}^{(n)}\right] \qquad \text{... (iii)}
\end{aligned}$$

The rate of convergence of the formula (*iii*) depends upon the choice of ω, which is called the accelerating factor and lies between 1 and 2.

Note: For $\omega = 1.875$, it was shown by B.A. Carr'e that the rate of convergence of (*iii*) is twice as great as when $\omega = 1$. For $\omega = 1.9$, the rate of convergence is 40 times greater as when $\omega = 1$. As a matter of fact, it is very difficult to estimate the best value of ω.

SOLVED EXAMPLES

Example 11.5. Solve the partial differential equation ; (Laplace's equation) $(\partial^2 u/\partial x^2) + (\partial^2 u/\partial y^2) = 0$ for the figure given below:

Solution: Using diagonal five-point formula; *viz.*

$$u_{i,j} = \frac{1}{4}\left[u_{i-1,j-1} + u_{i+1,j-1} + u_{i+1,j+1} + u_{i-1,j+1}\right]:$$

we obtain

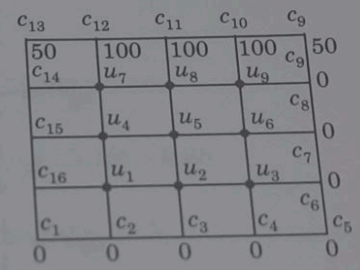
$$u_5 = \frac{1}{4}(c_1 + c_5 + c_9 + c_{13}) = \frac{1}{4}(0 + 0 + 50 + 50) = \frac{100}{4} = 25,$$

$$u_7 = \frac{1}{4}(c_{15} + u_5 + c_{11} + c_{13}) = \frac{1}{4}(0 + 25 + 100 + 50) = 43.75,$$

$$u_9 = \frac{1}{4}(u_5 + c_7 + c_9 + c_{11}) = \frac{1}{4}(25 + 0 + 50 + 100) = 43.75,$$

$$u_1 = \frac{1}{4}(c_1 + c_3 + u_5 + c_{15}) = \frac{1}{4}(0 + 0 + 25 + 0) = 6.25,$$

$$u_3 = \frac{1}{4}(c_3 + c_5 + c_7 + u_5) = \frac{1}{4}(0 + 0 + 0 + 25) = 6.25.$$



Then, the remaining quantities u_8, u_4, u_6, u_2 are calculated by the standard point formula and thus we get

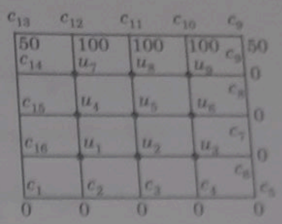
$$u_8 = \frac{1}{4}(u_5 + u_9 + u_{11} + u_7) = \frac{1}{4}(25 + 43.75 + 100 + 43.75) = 53.12,$$

$$u_4 = \frac{1}{4}(u_1 + u_5 + u_7 + c_{15}) = \frac{1}{4}(6.25 + 25 + 43.75 + 0) = 18.75,$$

$$u_6 = \frac{1}{4}(u_3 + c_7 + u_9 + u_5) = \frac{1}{4}(6.25 + 0 + 43.75 + 25) = 18.75,$$

$$u_5 = \frac{1}{4}(c_1 + c_5 + c_9 + c_{13}) = \frac{1}{4}(0 + 0 + 50 + 50) = \frac{100}{4} = 25,$$

$$u_7 = \frac{1}{4}(c_{15} + u_5 + c_{11} + c_{13}) = \frac{1}{4}(0 + 25 + 100 + 50) = 43.75,$$



$$u_9 = \frac{1}{4}(u_5 + c_7 + c_9 + c_{11}) = \frac{1}{4}(25 + 0 + 50 + 100) = 43.75,$$

$$u_1 = \frac{1}{4}(c_1 + c_3 + u_5 + c_{15}) = \frac{1}{4}(0 + 0 + 25 + 0) = 6.25,$$

$$u_3 = \frac{1}{4}(c_3 + c_5 + c_7 + u_5) = \frac{1}{4}(0 + 0 + 0 + 25) = 6.25.$$

Then, the remaining quantities

u_8, u_4, u_6, u_2 are calculated by the standard point formula and thus we get

$$u_8 = \frac{1}{4}(u_5 + u_9 + u_{11} + u_7) = \frac{1}{4}(25 + 43.75 + 100 + 43.75) = 53.12,$$

$$u_4 = \frac{1}{4}(u_1 + u_5 + u_7 + c_{15}) = \frac{1}{4}(6.25 + 25 + 43.75 + 0) = 18.75,$$

$$u_6 = \frac{1}{4}(u_3 + c_7 + u_9 + u_5) = \frac{1}{4}(6.25 + 0 + 43.75 + 25) = 18.75,$$

$$u_2 = \frac{1}{4}(c_3 + u_3 + u_5 + u_1) = \frac{1}{4}(0 + 6.25 + 25 + 6.25) = 9.38.$$

Thus all u_i,s ($i = 1, 2, 3, ...9$) are calculated, their accuracy is then improved, say by **Gauss-Seidel method.** First approximations of all the nine mesh points are given by

$$u_5^{(1)} = 25.00,\ u_7^{(1)} = 43.75,\ u_9^{(1)} = 43.75,\ u_1^{(1)} = 6.25,\ u_3^{(1)} = 6.25$$

$$u_8^{(1)} = 53.12,\ u_4^{(1)} = 18.75,\ u_6^{(1)} = 18.75,\ u_2^{(1)} = 9.38.$$

Now G.S. method gives

$$u_{i,j}^{(n+1)} = \frac{1}{4}\left[u_{i-1,j}^{(n+1)} + u_{i+1,j}^{(n)} + u_{i,j-1}^{(n+1)} + u_{i,j+1}^{(n)}\right]$$

$$u_1^{(2)} = \frac{1}{4}\left[c_{16} + u_2^{(1)} + c_2 + u_4^{(1)}\right] = \frac{1}{4}(0 + 9.38 + 0 + 18.75) = 7.03$$

$$u_2^{(2)} = \frac{1}{4}\left(u_1^{(2)} + u_3^{(1)} + c_3 + u_5^{(1)}\right) = \frac{1}{4}(7.03 + 6.25 + 0 + 25) = 9.57$$

$$u_3^{(2)} = \frac{1}{4}\left(u_2^{(2)} + c_6 + c_4 + u_6^{(1)}\right) = \frac{1}{4}(9.57 + 0 + 0 + 18.75) = 7.08$$

$$u_4^{(2)} = \frac{1}{4}\left(c_{15} + u_5^{(1)} + u_1^{(2)} + u_7^{(1)}\right) = 18.94$$

$$u_5^{(2)} = \frac{1}{4}\left(u_4^{(2)} + u_6^{(1)} + u_2^{(2)} + u_8^{(1)}\right)$$

$$= \frac{1}{4}(18.94 + 18.75 + 9.57 + 53.12) = (100.38/4) = 2.10$$

$$u_6^{(2)} = \frac{1}{4}\left(u_5^{(2)} + c_7 + u_3^{(2)} + u_9^{(1)}\right)$$

$$= \frac{1}{4}(25.10 + 0 + 7.08 + 43.75) = 18.98$$

$$u_7^{(2)} = \frac{1}{4}\left(c_{14} + u_8^{(1)} + u_4^{(2)} + c_{12}\right)$$

$$= \frac{1}{4}(0 + 53.12 + 18.94 + 100) = \frac{172.06}{4} = 43.02$$

$$u_8^{(2)} = \frac{1}{4}\left(u_7^{(2)} + u_9^{(1)} + c_{11}\right) = \frac{1}{4}(43.02 + 43.75 + 25.10 + 100) = 52.97$$

$$u_9^{(2)} = \frac{1}{4}\left(u_7^{(2)} + c_8 + u_6^{(2)} + c_{10}\right) = \frac{1}{4}(52.97 + 0 + 18.98 + 100)$$

$$= \frac{171.95}{4} = 42.99$$

These are the first iterations due to G.S. method

Further, we have

$$u_{i,j} = \frac{1}{4}\left[u_{i-1,j}^{(3)} + u_{i+1,j}^{(2)} + u_{i,j-1}^{(3)} + u_{i,j+1}^{(2)}\right]$$

$$u_1^{(3)} = \frac{1}{4}\left(c_{16} + u_2^{(2)} + c_2 + u_4^{(2)}\right) = \frac{1}{4}(0 + 9.57 + 0 + 18.94) = 7.13,$$

$$u_2^{(3)} = \frac{1}{4}\left[u_1^{(3)} + u_3^{(2)} + c_2 + u_4^{(2)}\right] = \frac{1}{4}(7.13 + 7.08 + 0 + 25.10) = 9.83,$$

$$u_3^{(3)} = \frac{1}{4}\left[u_2^{(3)} + c_6 + c_4 + u_6^{(2)}\right] = \frac{1}{4}(9.83 + 0 + 0 + 18.98) = 7.20,$$

$$u_4^{(3)} = \frac{1}{4}\left[c_{15} + u_5^{(2)} + u_1^{(3)} + u_7^{(2)}\right] = \frac{1}{4}(0 + 25.10 + 7.13 + 43.02) = 18.81,$$

$$u_5^{(3)} = \frac{1}{4}\left[u_4^{(3)} + u_6^{(2)} + u_2^{(3)} + u_8^{(2)}\right] = \frac{1}{4}(18.81 + 18.89 + 9.83 + 52.97) = 25.15$$

$$u_6^{(3)} = \frac{1}{4}\left[u_5^{(3)} + c_7 + u_3^{(3)} + u_9^{(2)}\right] = \frac{1}{4}(25.15 + 0 + 7.20 + 42.99) = 18.84,$$

$$u_7^{(3)} = \frac{1}{4}\left[c_{14} + u_8^{(2)} + u_4^{(3)} + c_{12}\right] = \frac{1}{4}(0 + 52.97 + 18.81 + 100) = 25.04,$$

$$u_8^{(3)} = \frac{1}{4}\left[u_7^{(3)} + u_9^{(2)} + u_5^{(3)} + c_{11}\right] = \frac{1}{4}(42.94 + 42.99 + 25.15 + 100) = 52.77,$$

$$u_9^{(3)} = \frac{1}{4}\left[u_8^{(3)} + c_8 + u_6^{(3)} + c_{10}\right] = \frac{1}{4}(52.77 + 0 + 18.84 + 100) = 42.90.$$

These give the second iterations due to G.S. method. To find the third iterations due to G.S. method, we again consider:

$$u_{i,j}^4 = \frac{1}{4}\left[u_{i-1,j}^{(4)} + u_{i+1,j}^{(3)} + u_{i,j-1}^{(4)} + u_{i,j+1}^{(3)}\right]$$

$$u_1^{(4)} = \frac{1}{4}\left(c_{16} + u_2^{(3)} + c_2 + u_4^{(3)}\right) = \frac{1}{4}(0 + 9.83 + 0 + 18.81) = 7.16,$$

$$u_2^{(4)} = \frac{1}{4}\left(u_1^{(4)} + u_3^{(3)} + c_3 + u_5^{(3)}\right) = \frac{1}{4}(7.16 + 7.20 + 0 + 25.15) = 9.88,$$

$$u_3^{(4)} = \frac{1}{4}\left(u_2^{(4)} + c_6 + c_4 + u_6^{(3)}\right) = \frac{1}{4}(9.88 + 0 + 0 + 18.84) = 7.18,$$

$$u_4^{(4)} = \frac{1}{4}\left(c_{15} + u_5^{(3)} + u_1^{(4)} + u_7^{(3)}\right) = \frac{1}{4}(0 + 25.15 + 7.16 + 42.94) = 18.81,$$

$$u_5^{(4)} = \frac{1}{4}\left(u_4^{(4)} + u_6^{(3)} + u_2^{(4)} + u_8^{(3)}\right) = \frac{1}{4}(18.81 + 18.84 + 9.88 + 52.77) = 25.08,$$

$$u_6^{(4)} = \frac{1}{4}\left(u_5^{(4)} + c_7 + u_3^{(4)} + u_9^{(3)}\right) = \frac{1}{4}(25.08 + 0 + 7.18 + 42.90) = 18.79,$$

$$u_7^{(4)} = \frac{1}{4}\left(c_{14} + u_8^{(4)} + u_4^{(4)} + c_{12}\right) = \frac{1}{4}\ 0 + 52.77 + 18.81 + 100) = 42.89,$$

$$u_8^{(4)} = \frac{1}{4}\left(u_7^{(4)} + u_9^{(3)} + u_5^{(4)} + c_{11}\right) = \frac{1}{4}(42.89 + 42.90 + 25.08 + 100) = 52.72,$$

$$u_9^{(4)} = \frac{1}{4}\left(u_8^{(4)} + c_8 + u_6^{(4)} + c_{10}\right) = \frac{1}{4}(52.72 + 0 + 18.79 + 100) = 42.88,$$

and the fourth iterations due to **G.S. method** are given as

$$u_1^{(5)} = \frac{1}{4}\left(c_{16} + u_2^{(4)} + c_4 + u_4^{(4)}\right) = \frac{1}{4}(0 + 9.88 + 0 + 18.81) = 7.17,$$

$$u_2^{(5)} = \frac{1}{4}\left(u_1^{(5)} + u_3^{(4)} + c_3 + u_5^{(4)}\right) = \frac{1}{4}(7.17 + 7.18 + 0 + 25.08) = 9.86,$$

$$u_3^{(5)} = \frac{1}{4}\left(u_2^{(5)} + c_6 + c_4 + u_6^{(4)}\right) = \frac{1}{4}(9.86 + 0 + 0 + 18.79) = 7.16,$$

$$u_4^{(5)} = \frac{1}{4}\left(c_{15} + u_5^{(4)} + u_1^{(5)} + u_7^{(4)}\right) = \frac{1}{4}(0 + 25.08 + 7.17 + 42.89) = 18.78,$$

$$u_5^{(5)} = \frac{1}{4}\left(u_4^{(5)} + u_6^{(4)} + u_2^{(5)} + u_8^{(4)}\right) = \frac{1}{4}(18.78 + 18.79 + 9.86 + 52.72) = 25.04$$

$$u_6^{(5)} = \frac{1}{4}\left(u_5^{(5)} + c_7 + u_3^{(5)} + u_9^{(4)}\right) = \frac{1}{4}(25.04 + 0 + 7.16 + 42.88) = 18.77,$$

$$u_7^{(5)} = \frac{1}{4}\left(c_{14} + u_8^{(4)} + u_4^{(5)} + c_{12}\right) = \frac{1}{4}(0 + 52.72 + 18.78 + 100) = 42.88,$$

$$u_8^{(5)} = \frac{1}{4}\left(u_7^{(5)} + u_9^{(4)} + u_5^{(5)} + c_{11}\right) = \frac{1}{4}(42.88 + 42.88 + 25.04 + 100) = 52.70,$$

$$u_9^{(5)} = \frac{1}{4}\left(u_8^{(5)} + c_8 + u_6^{(5)} + c_{10}\right) = \frac{1}{4}(52.70 + 0 + 18.77 + 100) = 42.87,$$

The next iterations are:

$$u_1^{(6)} = \frac{1}{4}\left(c_{16} + u_2^{(5)} + c_{12} + u_4^{(5)}\right) = \frac{1}{4}(0 + 9.86 + 0 + 18.78) = 7.16,$$

$$u_2^{(6)} = \frac{1}{4}\left(u_1^{(6)} + u_3^{(5)} + c_3 + u_5^{(5)}\right) = \frac{1}{4}(7.16 + 7.16 + 0 + 25.04) = 9.84,$$

$$u_3^{(6)} = \frac{1}{4}\left(u_2^{(6)} + c_6 + c_4 + u_6^{(5)}\right) = \frac{1}{4}(9.84 + 0 + 0 + 18.77) = 7.15,$$

$$u_4^{(6)} = \frac{1}{4}\left(c_{15} + u_5^{(5)} + u_1^{(6)} + u_7^{(5)}\right) = \frac{1}{4}(0 + 25.04 + 7.16 + 42.88) = 18.77,$$

$$u_5^{(6)} = \frac{1}{4}\left(u_4^{(6)} + u_6^{(5)} + u_2^{(6)} + u_8^{(5)}\right) = \frac{1}{4}(18.77 + 18.77 + 9.84 + 52.70) = 25.02$$

$$u_6^{(6)} = \frac{1}{4}\left(u_4^{(6)} + u_6^{(5)} + u_2^{(6)} + u_8^{(5)}\right) = \frac{1}{4}(25.02 + 0 + 7.15 + 42.87) = 18.76,$$

$$u_7^{(6)} = \frac{1}{4}\left(c_{14} + u_8^{(5)} + u_4^{(6)} + c_{12}\right) = \frac{1}{4}(0 + 52.70 + 18.77 + 100) = 42.87,$$

$$u_8^{(6)} = \frac{1}{4}\left(u_7^{(6)} + u_9^{(5)} + u_5^{(6)} + c_{11}\right) = \frac{1}{4}(42.87 + 42.87 + 25.02 + 100) = 52.69,$$

$$u_9^{(6)} = \frac{1}{4}\left(u_8^{(6)} + c_8 + u_6^{(6)} + c_{10}\right) = \frac{1}{4}(52.69 + 0 + 18.76 + 100) = 42.86.$$

All the above iterations are represented in tabular form below:

u_1	u_2	u_3	u_4	u_5	u_6	u_7	u_8	u_9
7.03	9.57	7.08	18.94	25.10	18.98	43.02	52.97	42.99
7.13	9.83	7.20	18.81	25.15	18.84	42.94	52.77	42.90
7.16	9.88	7.18	18.81	25.08	18.79	42.89	52.72	42.88
7.17	9.86	7.16	18.78	25.04	18.77	42.88	52.70	42.87
7.16	9.84	7.15	18.77	25.02	18.76	42.87	52.69	42.86

Example 11.6. Solve the partial difference equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ for figure given below, by (*i*) Jacobi's method, (*ii*) Gauss-Seidel's method, (*iii*) S.O.R. method. (Elliptic Type)

Solution: The approximate values of u_1, u_2, u_3, u_4 are calculated as follows:

$$u_1 = \frac{1}{4}(0 + 0 + 0 + 1) = 0.25,$$

2181 = .12464,

−.037216 = .37460,

−.037341 = 0.37495

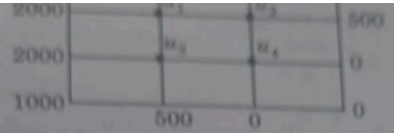
2413 = .12488

= .12499,

3746 = .37499

3749 = 0.375.

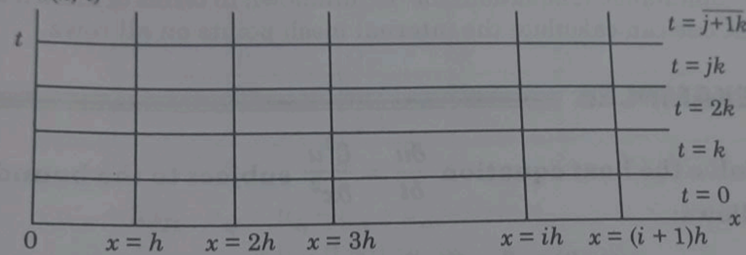
(a) and (b):



[**Ans.** $u_1 = 1208$, $u_2 = 792$, $u_3 = 1042$ and $u_4 = 458$]

11.4 PARABOLIC EQUATIONS

Consider, the heat conduction equation ; *viz.* $\dfrac{\partial u}{\partial t} = \dfrac{\partial^2 u}{\partial x^2}$ where $u = u(x, t)$ ... (i)



Let us divide the $x - t$ plane into a number of smaller rectangles by means of the lines $x = 0, x = h, x = 2h$ (*i.e.* $x = ih; i = 0, 1, 2, ...$) and $t = 0, k, 2k, ...$ (*i.e.* $t = jk ; j = 0, 1, 2, ...$). Then we readily obtain $\dfrac{\partial^2 u}{\partial x^2} = \dfrac{1}{h^2}\,[u_{i-1,j} - 2u_{i,j} + u_{i+1,j}]$ and $\dfrac{\partial u}{\partial t} = \dfrac{1}{k}\,[u_{i,j+1} - u_{i,j}]$. Putting these values in (*i*), we get

$$\frac{1}{k}[u_{i,j+1} - u_{i,j}] = \frac{1}{h^2}[u_{i-1,j} - 2u_{i,j} + u_{i+1,j}]$$

$$\Rightarrow \quad u_{i,j+1} - u_{i,j} = r(u_{i-1,j} - u_{i,j} + u_{i+1,j}) \text{ where } r = (k/h^2)$$

$$\Rightarrow \quad u_{i,j+1} = u_{i,j} + r[u_{i-1,j} - 2u_{i,j} + u_{i+1,j}] \quad \text{... (ii)}$$

This formula (*ii*) gives the value of the unknown function at the $(i, j+1)$th interior point in terms of the known functional values and hence is known as *explicit formula* and remains valid only for $0 < r \le \left(\frac{1}{2}\right)$. It is important to note here that for the approximation of $\left(\frac{\partial^2 u}{\partial x^2}\right)$, we have used the functional value along the jth row only. In 1947 **CRANK** and **NICOLSON** suggested the method according to which $\left(\frac{\partial^2 u}{\partial x^2}\right)$ is replaced by the average value of its finite-difference approximations on the jth and $(j+1)$th rows implying.

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{2}\left[\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2} + \frac{u_{i-1,j+1} - 2u_{j+1,j} + u_{i+1,j+1}}{h^2}\right]$$

Now putting this value of $\frac{\partial^2 u}{\partial x^2}$ and $\frac{\partial u}{\partial t}$ in equation (*i*), we get

$$\frac{1}{k}\{u_{i,j+1} - u_{i,j}\} = \frac{1}{2h^2}\{u_{i-1,j} - 2u_{i,j} + u_{i+1,j} + u_{1-1,j+1} - 2u_{i,j+1} + u_{i+1,j+1}\}$$

$$\Rightarrow \quad -ru_{i-1,j+1} + (2+2r)u_{i,j+1} - ru_{i+1,j+1}$$
$$= ru_{i-1,j} + 2(1-r)u_{i,j+1} + ru_{i+1,j}, \text{ where } r = (k/h)^2 \quad \text{... (iii)}$$

On the L.H.S. of this equation we have three unknowns while on R.H.S. all the three are known. The implicit formula (*iii*) is known as *Crank-Nicolson formula* and remains convergent for all finite values of r. If in each row, we have "m" internal mesh points then the above formula gives "m" simultaneous equations for "m" unknown in terms of known boundary values. On the same lines, one can calculate the internal mesh points on all rows.

SOLVED EXAMPLES

Example 11.7. Solve the heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ subject to the boundary and initial conditions as follows:

$$u(x, 0) = 0,\ u(0, t) = 0,\ u(1, t) = t.$$

Solution: By 11.4 relation (*iii*), we have

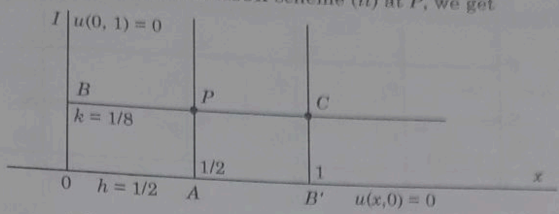
$$= ru_{i-1,j+1} + (2+2r)u_{i,j+1} + ru_{i+1,j+1}$$
$$= ru_{i-1,j} + 2(1-r)u_{i,j} + ru_{i+1,j} \text{ where } r = (k/h)^2$$

and $\quad u \equiv u(x, t) \quad \text{... (i)}$

(A) Let us choose $h = \frac{1}{2}$ and $k = \frac{1}{8}$ so that $r = \frac{1}{2}$. Then (*i*) can be re-written as

$$-u_{i-1,j+1}+6u_{i,j+1}-u_{i+1,j+1}=u_{i-1,j}+2u_{i,j}+u_{i+1,j} \quad \text{... (ii)}$$

Let $OA = h = \frac{1}{2}$, $OB = k = \frac{1}{8}$. Then co-ordinates of P are given by $\left(\frac{1}{2},\frac{1}{8}\right)$. Let the value of u at P be u_1. Now applying **Crank-Nicolson** scheme (*ii*) at P, we get



$$-u_B+6u_P-u_C=u_O+2u_A+u'_B$$

$$\Rightarrow \quad -0+6u_1-u\left(1,\frac{1}{8}\right)=0+2u\left(\frac{1}{2},0\right)+u(1,0)$$

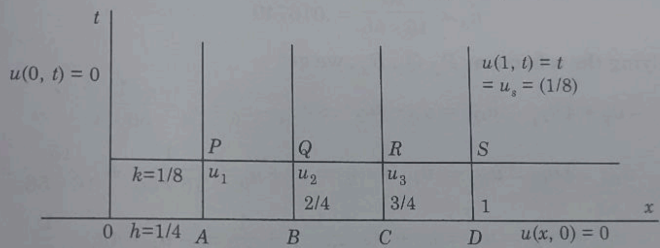
$$\Rightarrow \quad -0+6u_1-\frac{1}{8}=0+2.0+0 \qquad (\because u(x,0)=0)$$

$$\Rightarrow \quad 6u_1=\frac{1}{8} \Rightarrow u_1=(1.48)=0.02083.$$

(B) Let us choose $k=\frac{1}{8}$ and $h=\frac{1}{4}$ so that $r=\left(\frac{k}{h^2}\right)=2$. It is worth noticing that C-N scheme is convergent for all finite values of r. For $r=2$, equation (*i*) implies

$$-u_{i-1,j+1}+3u_{i,j+1}-u_{i+1,j+1}=u_{i-1,j}-u_{i,j}+u_{i+1,j} \quad \text{... (iii)}$$

Applying this equation for the mesh point P in the diagram, we have



$$-u'_P+3u_P-u_Q=u_Q-u_A+u_B \text{ where } u_{i,j+1}\equiv u_1$$

$$\Rightarrow \quad -0+3u_1-u_2=0-0+0 \Rightarrow 3u_1=u_2$$

Again applying the same scheme at Q and R respectively, we obtain

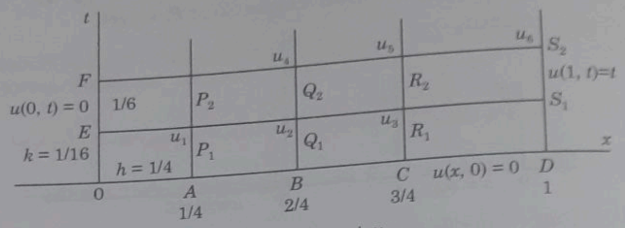
$$-u_P+3u_Q-u_R=u_A-u_B+u_C=0 \Rightarrow -u_1+3u_2-u_3=0$$

and

$$-u_Q+3u_R-u_S=u_B-u_C+u_D=0 \Rightarrow -u_2+3u_3-\frac{1}{8}=0.$$

Thus we have three unknown and three equations. Solving these, one may get $u_1 = .00595$, $u_2 = 0.01785$ and $u_3 = 0.04760$.

(C) We now consider $k = \frac{1}{16}$ and $h = \frac{1}{4}$ so that $r = 1$. For $r = 1$, C-N scheme can be written as



$$-u_{i-1,j+1} + 4u_{i,j+1} - u_{i+1,j+1} = u_{i-1,j} + u_{i,j} + u_{i+1,j}$$

Applying this scheme at P_1, Q_1, R_1, we get

$$-u_E + 4u_{P_1} - u_{Q_1} = u_O + u_B \text{ where } u_{i,j+1} \equiv u_1$$

i.e. $$-0 + 4u_1 - u_2 = 0 \quad \text{--- (1)}$$

$$-u_{P_1} - 4u_{Q_1} - u_{R_1} = u_A + u_C = 0 \Rightarrow -u_1 + 4u_2 - u_3 = 0 \quad \text{--- (2)}$$

and $$-u_{Q_1} - 4u_{R_1} - u_{S_1} = u_B + u_D = 0 \Rightarrow -u_2 + 4u_3 - u_{S_1} = 0 \quad \text{--- (3)}$$

But $$u_{S_1} = [t]_{t=(1/16)} = \frac{1}{16} \Rightarrow -u_2 + 4u_3 - \frac{1}{16} = 0 \quad \text{--- (4)}$$

Solving (1) and (2) and (4) for u_1, u_2, u_3; we get

$$u_1 = \frac{1}{16 \times 56} = .001116,\ u_2 = \frac{1}{56 \times 4} = .004464$$

and $$u_3 = \frac{15}{16 \times 56} = .016740$$

Again applying the scheme at P_2, Q_2, R_2 ; we get

$$-u_F + 4u_{P_2} - u_{Q_2} = u_E + u_{Q_1} \Rightarrow 4u_4 - u_5 = u_2 = \frac{1}{56 \times 4} \quad \text{--- (5)}$$

$$-u_{P_2} - 4u_{Q_2} - u_{R_2} = u_{P_1} + u_{R_1} = u_1 + u_3 = \frac{1}{16 \times 56} + \frac{15}{16 \times 56} = \frac{1}{56} \quad \text{--- (6)}$$

and $$-u_{Q_2} + 4u_{R_2} - u_{S_2} = u_2 + u_{S_1} \frac{1}{56 \times 4} + \frac{1}{16} \Rightarrow -u_5 + 4u_6 - \frac{1}{8}$$

$$= \frac{1}{56 \times 4} + \frac{1}{16} \quad \text{--- (7)}$$

$$\left(\because u_{S_2} = \frac{1}{8} \text{ and } u_{S_1} = \frac{1}{16}\right)$$

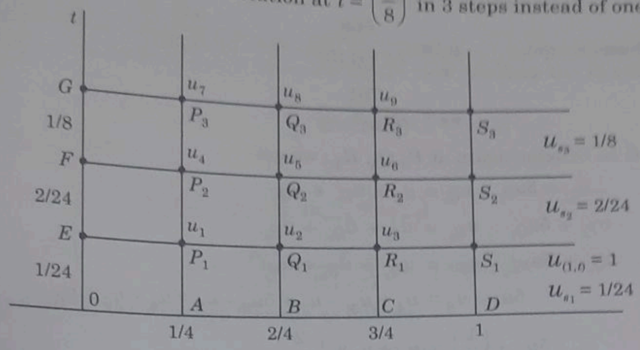
Now solving (5), (6), (7); we get

$$u_4 = .005899,\ u_5 = 0.019132 \text{ and } u_6 = 0.052771.$$

Note: Here we have obtained the solution for $t = 1/8$ in two steps instead of one step.

(D) As our final choice, we choose $k = \frac{1}{24}$, $h = \frac{1}{4}$ so that $r = \left(\frac{k}{h^2}\right) = \left(\frac{1}{24} / \frac{1}{16}\right) = \left(\frac{2}{3}\right)$.

Thus we now propose to find the solution at $t = \left(\frac{1}{8}\right)$ in 3 steps instead of one step.



For $\qquad r = \left(\frac{2}{3}\right)$, C-N scheme can be written as

$$-\frac{2}{3}\, u_{i-1,j+1} + \frac{10}{3}\, u_{i,j+1} - \frac{2}{3}\, u_{i+1,j}j + 1 = \frac{2}{3}\, u_{i-j}$$

$$+\frac{2}{3}\, u_{i+1,j}\, \frac{2}{3}\, u_{e+1,j}$$

$$\Rightarrow \quad -u_{i-1,j+1} + 5u_{i,j+1} - u_{i+1,j+1} = u_{i-1,j} + u_{i,j} + u_{i+1,j} \qquad \text{... (1)}$$

Applying this equation for the mesh points P_1, Q_1, R_1 respectively; we get

$$-u_E + 5U_{P_1} - u_{Q_1} = u_O = u_O + u_A + u_B \qquad \text{... (2)}$$

$$-u_{P_1} + 5U_{Q_1} - u_{R_1} = u_A + u_B + u_C \qquad \text{... (3)}$$

$$-u_{Q_1} + 5U_{R_1} - u_{S_1} = u_B + u_C + u_D \qquad \text{... (4)}$$

$$\Rightarrow \quad 5u_1 - u_2 = 0,\ -u_1 + 5u_2 - u_3 = 0,\ -u_2 + 5u_3 - \frac{1}{24} = 0$$

Solving these we get $\qquad u_1 = \frac{1}{115 \times 24},\ u_2 = \frac{1}{23 \times 24},\ u_3 = -\frac{1}{115}.$

Applying C.N. scheme at P_2, Q_2, R_2 ; we get

$$-u_F + 5U_{P_2} - u_{Q_2} = u_E + u_{P_1} + u_{Q_1}$$

$$-u_{P_2} + 5U_{Q_2} - u_{R_2} = u_{P_1} + u_{Q_1} + u_{R_1}$$

$$-u_{Q_2} + 5U_{R_2} - u_{S_2} = u_{Q_1} + u_{R_1} + u_{S_1}$$

$$\Rightarrow \quad -0 + 5u_4 - u_5 = u_1 + u_2 = \frac{1}{4 \times 115}$$

$$-u_4 + 5u_5 - u_6 = u_1 + u_2 + u_3 = \frac{1}{92},$$

and $$-u_5 + 5u_6 = \frac{2}{24} + u_2 + u_3 + \frac{1}{24} = \frac{374}{24 \times 115}$$

Solving these, we get $$u_4 = \frac{668}{24 \times (115)^2},\ u_5 = \frac{53}{12 \times (23)^2}$$

$$u_6 = \frac{761}{(115)^2 \times 2}.$$

Applying C.N. Scheme again at P_3, Q_3, R_3 ; we get

$$-u_G + 5u_{P_3} - u_{Q_3} = u_F + u_{P_2} + u_{Q_2}$$

$$-u_{P_3} + 5u_{Q_3} - u_{R_3} = u_{P_2} + u_{Q_2} + u_{R_2}$$

and $$-u_{Q_3} + 5u_{R_3} - u_{S_3} = u_{Q_2} + u_{R_2} + u_{S_2}$$

$\Rightarrow$ $$5u_7 - u_8 = u_4 + u_5,\ -u_7 + 5u_8 - u_9 = u_4 + u_5 + u_6$$

and $$-u_8 + 5u_9 - \left(\frac{1}{8}\right) = u_5 + u_6 + \left(\frac{2}{24}\right)$$

$\Rightarrow$ $$5u_7 - u_8 = \frac{553}{(115)^2 \times 4},\ -u_7 + 5u_8 - u_9 = \frac{83}{(23)^2 \times 4}$$ and

$$-u_8 + 5u_9 = \frac{77907}{(23)^2 \times 24 \times 25}$$

Solving, we obtain

$$u_7 = .0060214,\ u_8 = .019654 \text{ and } u_9 = .053022$$

It reads as $$u(x, t) = \frac{1}{6}(x^3 - x + 6xt) +$$

$$\frac{2}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^3} e^{-n^2\pi^2 t} \sin n \pi x$$

$\Rightarrow$ $$u\left(\frac{1}{4}, \frac{1}{8}\right) = 0.00541,\quad u\left(\frac{1}{2}, \frac{1}{8}\right) = 0.01878$$

and $$u\left(\frac{3}{4}, \frac{1}{8}\right) = .05240.$$

11.5 ITERATIVE METHODS

Iterative methods, discussed earlier can be used to solve finite difference equation:

$$-ru_{i-1,j+1} + (2 + 2r)\, u_{i,j+1}$$
$$-ru_{i+1,j+1} = r\, u_{i-1,j} + (2 - 2r)\, u_{i,j} + ru_{i+1,j}$$ where

$r = \left(\frac{k}{h^2}\right)$ **[Crank-Nicolson formula]** obtained by the heat conduction equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$.

$$\Rightarrow \qquad 5u_7 - u_8 = \frac{\quad}{(115)^2 \times 4}$$

$$-u_8 + 5u_9 = \frac{77907}{(23)^2 \times 24 \times 25}$$

Solving, we obtain

$$u_7 = .0060214,\ u_8 = .019654 \text{ and } u_9 = .053022$$

It reads as

$$u(x, t) = \frac{1}{6}(x^3 - x + 6xt) + \frac{2}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^3} e^{-n^2\pi^2 t} \sin n\pi x$$

$$\Rightarrow \qquad u\left(\frac{1}{4}, \frac{1}{8}\right) = 0.00541, \quad u\left(\frac{1}{2}, \frac{1}{8}\right) = 0.01878$$

and

$$u\left(\frac{3}{4}, \frac{1}{8}\right) = .05240.$$

11.5 ITERATIVE METHODS

Iterative methods, discussed earlier can be used to solve finite difference equation:

$$-ru_{i-1,j+1} + (2+2r)\,u_{i,j+1} - ru_{i+1,j+1} = r\,u_{i-1,j} + (2-2r)\,u_{i,j} + ru_{i+1,j} \text{ where}$$

$r = \left(\dfrac{k}{h^2}\right)$ **[Crank-Nicolson formula]** obtained by the heat conduction equation $\dfrac{\partial u}{\partial t} = \dfrac{\partial^2 u}{\partial x^2}$.

Jacobi's ... iterates only ... can be seen easily that while com... latest value of u_{i-1}, namely $u_i^{(n+1)}$ is known already. Thus the converge...

Iteration Formula can be improved by putting $u_{i-1}^{(n+1)}$, for $u_{i-1}^{(n)}$ and acc... formula

$$u_i^{(n+1)} = \frac{p_i}{1+r} + \frac{r}{2(1+r)}\left\{u_{i-1}^{(n+1)} + u_{i+1}^{(n)}\right\}$$

This formula (6) is known as **Gauss-Seidel iteration form**... finite value of r. *This formula converges twice as fast as Jacob*... also be written as

$$u_i^{(n+1)} = u_i^{(n)} + \left\{\frac{r}{2(1+r)}\left(u_{i-1}^{(n+1)} + \ldots\right.\right.$$

From (7), it is clear that the expression within the curl... the nth and $(n+1)$th iterates. If this difference is taken a... we have

$$u_i^{(n+1)} = u_i^{(n)} + \omega\left\{\frac{r}{2(1+r} \ldots\right.$$

This formula is known as *Successive-Over Rela*... called the relaxation factor and generally lies in b...

The Crank-Nicolson formula can be further written as

$$(1+r)u_{i,j+1} = u_{i,j} + \left(\frac{r}{2}\right)\{u_{i-1,j+1} + u_{i+1,j+1} + u_{i-1,j} - 2u_{i,j}\}$$

where

$$r = \left(\frac{k}{h^2}\right) \quad \text{... (1)}$$

In this equation, the unknowns are $u_{i,j+1}$, $u_{i-1,j+1}$ and $u_{i+1,j+1}$ and all others are known as these were computed at the jth step. Keeping this in view, we drop the j's and then setting

$$p_i = u_{i,j} + \frac{r}{2}\{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}\} \quad \text{... (2)}$$

in (1), we have obtained

$$(1+r)\,u_i = p_i + \frac{r}{2}\,(u_{i-1} + u_{i+1}) \quad \text{... (3)}$$

$$\Rightarrow \quad u_i = \frac{p_i}{1+r} + \frac{r}{2(1+r)}\{u_{i-1} + u_{i+1}\} \quad \text{... (4)}$$

From (4), we obtain the iterative formula as follows:

$$u_i^{(n+1)} = \frac{p_i}{1+r} + \frac{r}{2(1+r)}\left\{u_{i-1}^{(n)} + u_{i+1}^{(n)}\right\} \quad \text{... (5)}$$

Formula (5) gives the $(n+1)$th iterate in terms of the nth iterates only and is known as **Jacobi's iteration formula.** By (5), it can be seen easily that while computing $u_i^{(n+1)}$, the latest value of u_{i-1}, namely $u_i^{(n+1)}$ is known already. Thus the convergence of the **Jacobi's Iteration Formula** can be improved by putting $u_{i-1}^{(n+1)}$, for $u_{i-1}^{(n)}$ and accordingly we obtain the formula

$$u_i^{(n+1)} = \frac{p_i}{1+r} + \frac{r}{2(1+r)}\left\{u_{i-1}^{(n+1)} + u_{i+1}^{(n)}\right\} \quad \text{... (6)}$$

This formula (6) is known as **Gauss-Seidel iteration formula** and is convergent for all finite value of r. *This formula converges twice as fast* as **Jacobi's scheme**. Equation (6) can also be written as

$$u_i^{(n+1)} = u_i^{(n)} + \left\{\frac{r}{2(1+r)}\left(u_{i-1}^{(n+1)} + u_{i+1}^{(n)}\right) + \frac{p_i}{1+r} - u_i^{(n)}\right\} \quad \text{... (7)}$$

From (7), it is clear that the expression within the curly brackets is the difference between the nth and $(n+1)$th iterates. If this difference is taken as w times the above expression, then we have

$$u_i^{(n+1)} = u_i^{(n)} + \omega\left\{\frac{r}{2(1+r)}\left(u_{i-1}^{(n+1)} + u_{i+1}^{(n)}\right) + \frac{p_i}{1+r} - u_i^{(n)}\right\} \quad \text{... (8)}$$

This formula is known as *Successive-Over Relaxation method* (**S.O.R.** method) while ω is called the relaxation factor and generally lies in between 1 and 2.

...ollows: ... 79.2, $u_3 = 104.2$, $u_4 = 45.9$.

Iteration u_i	0	1	2	3	4	5
u_1	100	112.5	119.1	120.4	120.8	120.9
u_2	60.0	75.6	78.3	79.0	79.2	79.2
u_3	90.0	100.6	103.3	104.0	104.2	104.2
u_4	40	44.1	45.4	45.8	45.9	45.9

EXERCISE 11.3

1. Solve $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial^2 x}$ in $0 < x < 5$, $t > 0$ given that $u(x, 0) = 20$, $u(0, t) = 0$, $u(5, t) = 100$. Compute u for the time-step with $h = 1$ by Crank-Nicholson method. [Ans. 10.05, 20.2, 30.75, 62.68]
2. Solve the boundary value problem $u_t = u_{xx}$ under the condition $u(0, t) = u(1, t)$ and $u(x, 0) = \sin \pi x$, $0 \le x \le 1$ (Take $h = 0.2$, and $\alpha = 1/2$).

[Ans.

$t \downarrow$ / $x \rightarrow$		0	0.2	0.4	0.6	0.8	1.0
	$j \backslash i$	0	1	2	3	4	5
0	0	0	0.5878	0.9511	0.9511	0.5878	0
0.02	1	0	0.4756	0.7695	0.7695	0.4756	0
0.04	2	0	0.3848	0.6225	0.6225	0.3848	0
0.06	3	0	0.3113	0.5036	0.5036	0.3113	0
0.08	4	0	0.2518	0.4074	0.4074	0.2518	0
0.1	5	0	0.2039	0.3296	0.3296	0.2037	0

]

11.7 POISSON'S EQUATION

The partial differential equation $\nabla^2 u = F(x, y)$

i.e., $$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial y^2} = F(x, y) \quad \text{... (i)}$$

where $F(x, y)$ is a given function x and y, is known as Poisson's equation.

We can solve it numerically at the points of a square mesh, by replacing the derivatives by difference expressions at the points $x = ih$, $y = jk$ (here $k = h$), and then substituting in (i), we get

$$\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2} + \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{h^2} = F(ih, jh)$$

$$\Rightarrow \quad (u_{i-1,j} + u_{i+1,j}) + (u_{i,j-1} + u_{i,j+1}) - 4u_{i,j} = h^2 F(ih, jh) \quad \text{... (ii)}$$

Using replacement formula at each mesh point, we can obtain similar equations in i, j and then such equations can be solved by iteration techniques.

By replacing $\frac{\partial^2 u}{\partial x^2}$ with the difference expression we get an error of the order $O(h^2)$ and also there exists an error of order $O(h^2)$, when we replace $\frac{\partial^2 u}{\partial y^2}$ by its difference expression. Hence the error in solving Poisson's equation by difference methods is of the order $O(h^2)$.

SOLVED EXAMPLES

Example 11.11. **Solve the partial differential equation**

$$\nabla^2 u = -10(x^2 + y^2 + 10)$$

over the square with sides $x = 0 = y$, $x = 3 = y$ with $u = 0$ on the boundary and mesh length = 1.

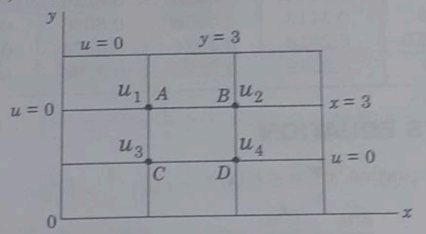
Solution: Let the values of u at the four mesh points be u_1, u_2, u_3, u_4 respectively. The differential equation is given by

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -10\,(x^2 + y^2 + 10) \quad \text{... }(i)$$

Now putting $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ in the form of finite difference expressions and putting $x = ih$, $y = jh$ $(h = 1)$ in (i), we get

$$u_{i-1,j} - 2u_{i,j} + u_{i+1,j} + u_{i,j-1} - 2u_{i,j} + 2u_{i,j+1} = -10\,(i^2 + j^2 + 10) \quad \text{... }(ii)$$

Using this formula at A, where $i - 1$ and $j = 2$, we get



$$0 + 0 + u_2 + u_3 - 4u_1 = -10\,(1 + 4 + 10)$$

$$\Rightarrow \quad u_2 + u_3 - 4u_1 = -150 \quad \text{... }(iii)$$

Also, applying the formula at B, where $i = 2$ and $i = 2$; we obtain

$$u_1 + 0 + 0 + u_4 - 4u_2 = -10\,(4 + 4 + 10)$$

$$\Rightarrow \quad u_1 + u_4 - 4u_2 = -180 \quad \text{... }(iv)$$

Again applying the formula at C, where $i = j$ and $j = 1$; we obtain

$$0 + u_1 + u_4 + 0 - 4u_3 = -10\,(1 + 1 + 10)$$

$$\Rightarrow \quad u_1 + u_4 - 4u_3 = -120 \quad \text{... }(v)$$

Finally, applying the formula at D, where $i = 2$ and $j = 1$; we get

$$u_3 + u_2 + 0 + 0 - 4u_2 = -10\,(4 + 1 + 10)$$

$$\Rightarrow \quad u_3 + u_2 - 4u_4 = -150 \quad \text{... }(vi)$$

We now solve equations (3) to (6) by Gauss-Seidel iteration technique (G.S. method). The process of iteration will converge, if in each equation of the system, the *abs* value of the largest coefficient is greater than the sum of the absolute values of all the remaining coefficients.

This condition holds good in equations (3) to (6).

Now writing the equations in the form:

$$\frac{310}{4} = 77.8125$$

$$u_1 = \frac{1}{4}(u_2 + u_3 + 150) \qquad \text{... (vii)}$$

$$u_2 = \frac{1}{4}(u_1 + u_4 + 180) \qquad \text{... (viii)}$$

$$u_3 = \frac{1}{4}(u_1 + u_4 + 120) \qquad \text{... (ix)}$$

$$u_4 = \frac{1}{4}(u_2 + u_3 + 150) \qquad \text{... (x)}$$

we obtain $u_4 = u_1$. So it is enough if we can find u_1, u_2, u_3.

Now, let us start the iteration by putting $u_2 = .0 = u_3$ in (7).

$$\Rightarrow \qquad u_1^{(1)} = \frac{150}{4} = 37.5 = u_4^{(1)}$$

Putting $u_1 = 37.5 = u_4^{(1)}$ in (8) and (9), we obtain

$$u_2^{(1)} = \frac{1}{1}(75 + 180) = \frac{255}{4} = 63.75$$

and

$$u_3^{(1)} = \frac{1}{4}(75 + 120) = \frac{145}{4} = 48.75$$

For the second iteration, we write

$$u_4^{(2)} = u_1^{(2)} = \frac{1}{4}(63.75 + 48.75 + 150) = \frac{262.5}{4} = 65.6250$$

$$u_{(2)}^{(2)} = \frac{1}{4}(65.625 + (-5.625 + 180)$$

$$u_3^{(2)} = \frac{1}{4}(65.625 + 65.625 + 120)$$

$$= \frac{250}{4} = 62.8125$$

Proceeding for this iteration, we get

$$u_4^{(3)} = u_1^{(3)} = \frac{1}{4}(77.8125 + 62.8125 + 150) = 72.65.6563$$

$$u_2^{(3)} = \frac{1}{4}(72.6563 + 72.6563 + 180) = 81.3282$$

$$u_3^{(3)} = \frac{1}{4}(72.6563 + 72.6563 + 120) = 66.3282$$

and then for the fourth iteration, we get

$$u_4^{(4)} = u_1^{(4)} = \frac{1}{4}(81.3282 + 66.3282 + 150) = 74.4141$$

$$u_2^{(4)} = \frac{1}{4}(74.4141 + 74.4141 + 180) = 82.2071$$

$$u_3^{(4)} = \frac{1}{4}(74.4141 + 74.4141 + 120) = 67.2071$$

For the fifth iteration, we obtain

$$u_1^{(5)} = \frac{1}{4}(82.2071 + 67.2071 + 150) = 74.8536 = u_4^{(5)}$$

$$u_2^{(5)} = \frac{1}{4}(74.8536 + 74.8536 + 180) = 82.4268$$

$$u_3^{(5)} = \frac{1}{4}(74.8536 + 74.8536 + 120) = 67.4268$$

For the sixth iteration, we have

$$u_1^{(6)} = u_4^{(6)} = \frac{1}{4}\left(u_2^{(5)} + u_3^{(5)} + 150\right)$$

$$= \frac{1}{4}(82.4268 + 67.4268 + 150) = 74.9634$$

$$u_2^{(6)} = \frac{1}{4}(74.8536 + 74.8536 + 180) = 82.4268$$

and $$u_3^{(6)} = \frac{1}{4}(74.8536 + 74.8536 + 150) = 67.4268$$

For the seventh iteration, we have

$$u_1^{(7)} = u_4^{(7)} = \frac{1}{4}\left(u_2^{(6)} + u_3^{(6)} + 150\right) = \frac{1}{4}(82.4268 + 67.4268 + 150) = 74.9634$$

$$u_2^{(7)} = \frac{1}{4}\left(u_1^{(6)} + u_4^{(6)} + 180\right) = \frac{1}{4}(74.9634 \times 2 + 180) = 82.4817$$

$$u_3^{(7)} = \frac{1}{4}\left(u_1^{(6)} + u_4^{(6)} + 120\right) = \frac{1}{4}(74.9634 \times 2 + 120) = 67.4817$$

For the eighth iteration, we have

$$u_1^{(8)} = u_4^{(8)} = \frac{1}{4}\left(u_2^{(7)} + u_3^{(7)} + 150\right) = \frac{1}{4}(82.4817 + 67.4817 + 150) = 74.99085$$

$$u_2^{(8)} = \frac{1}{4}\left(2u_1^{(7)} + 180\right) = \frac{1}{4}(2 \times 74.9634 + 180) = 82.4817$$

$$u_3^{(8)} = \frac{1}{4}\left(2u_1^{(7)} + 120\right) = \frac{1}{4}(2 \times 74.9634 + 120) = 67.4817$$

For the ninth iteration, we have

$$u_1^{(9)} = u_4^{(9)} = \frac{1}{4}\left(u_2^{(8)} + u_3^{(8)} + 150\right) = \frac{1}{4}(82.4817 + 67.4817 + 150) = 74.99085$$

$$u_2^{(9)} = \frac{1}{4}\left(2u_1^{(8)} + 180\right) = \frac{1}{4}(2 \times 74.99085 + 180) = 82.4954$$

$$u_3^{(9)} = \frac{1}{4}\left(2u_1^{(8)} + 120\right) = \frac{1}{4}(2 \times 74.99085 + 120) = 67.4954$$

For the tenth iteration, we have

$$u_3^{(10)} = u_1^{(10)} = \frac{1}{4}\left(u_2^{(9)} + u_3^{(9)} + 150\right) = \frac{1}{4}(82.454 + 67.4954 + 150) = 74.9977$$

$$u_2^{(10)} = \frac{1}{4}\left(2u_1^{(9)} + 180\right) = \frac{1}{4}(2 \times 74.99085 + 180) = 82.4954$$

$$u_3^{(10)} = \frac{1}{4}\left(2u_1^{(9)} + 120\right) = \frac{1}{4}(2 \times 74.990 + 120) = 67.4954$$

Hence $u_1 = u_4 = 74.99$, $u_2 = 82.49$, $u_3 = 67.49$; correct to two place of decimal.

11.8 PARABOLIC EQUATIONS (SOLUTION BY BENDER SCHMIDT RECURRENCE RELATION)

One dimensional heat equation is given by

$$\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2} \text{ where } \alpha^2 = \frac{k}{c\rho},$$ where c is the specific heat of the material, ρ is the density and k is the thermal conductivity. This implies

$$u_t = \alpha^2 u_{xx} \text{ or } u_{xx} - au_t = 0 \text{ where } a = \left(\frac{1}{\alpha^2}\right). \quad \text{... } (i)$$

Here $A = 1$, $B = 0$, $C = 0$ and hence, we have

$$\Delta S = B_2 - 4\,AC = 0 - 0 = 0 \Rightarrow \text{the equation is parabolic.}$$

Let us solve it by the method of finite difference, under the boundary conditions:

$$u(0, t) = T_0 \quad \text{... } (ii)$$

$$u(l, t) = T_l \quad \text{... } (iii)$$

and initial condition: $u(x, 0) = f(x)$

We have $u_{xx} = \dfrac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2}$, $u_t = \dfrac{u_{i,j+1} - u_{i,j}}{k}$ putting these values in (i), we have

$$\frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2} - a \cdot \left(\frac{u_{i,j+1} - u_{i,j}}{k}\right) = 0$$

$$\Rightarrow \quad u_{i,j+1} - u_{i,j} = \frac{k}{h^2 a}(u_{i+1,j} + u_{i-1,j})$$

$$= \lambda(u_{i+1,j} - 2u_{i,j} + u_{i-1,j}) \text{ where } \lambda = \frac{k}{h^2 a}$$

$$\Rightarrow \quad u_{i,j+1} = \lambda u_{i+1,j} + (1 - 2\lambda)\,u_{i,j} + \lambda u_{i-1,j} \quad \text{... } (iv)$$

The boundary conditions (ii) can be put in different form as

$$\left.\begin{aligned} u_{0,j} &= T_0 \\ u_{n,j} &= T_l \end{aligned}\right\} j = 1, 2, \ldots \quad \text{... } (v) \text{ where } nh = l$$

The initial condition (iii) can be written as

$$u_{i,0} = f(ih)\; i = 1, 2, \ldots \quad \text{... } (vi)$$

Now equation (iv) gives the value of u at $x = ih$, and $t_j + k$ in terms of values of u at $x = (i-1)h$, ih and $x = (i+1)h$ at a time t_j.

But $\quad u(x, 0) = f(x)$,

hence the recurrence relation (*iv*) gives u at each pivotal point x_i at any time t_j. Furthermore, the equation (*iv*) becomes simple, if for a given h, k; we choose $1 - 2\lambda = 0 \Rightarrow \lambda = \frac{1}{2}$, *i.e.*, $\frac{k}{h^2 a} = \frac{1}{2}$ or $k = \frac{h^2 a}{2}$.

Now equation (*iv*) can be rewritten as

$$u_{i,j+1} = \frac{1}{2}(u_{i+1,j} + u_{i-1,j}) \qquad \text{... (vii)}$$

i.e., u at $x = x_i$ at a time t_{j+1} is equal to the average of the values of u at the surrounding points x_{i+1} and x_{i-1} at the previous time t_j *I*. Relation (*vii*) is called the **Bender-Schmidt recurrence relation.**

SOLVED EXAMPLES

Example 11.12. Find the solution to the parabolic equation $\frac{\partial^2 u}{\partial x^2} - 2\frac{\partial u}{\partial t} = 0$ where $u(0, t) = 0$, $u(4, t) = 0$, $u(x, 0) = x(4 - x)$. Assume $h = 1$. Find the values upto $t = 5$.

Solution: The general heat equation is $u_{xx} = a \cdot u_t$ ... (*i*)

Here, $a = 2$, $h = 1$; $\quad \lambda = \frac{h}{h^2 a} = \frac{1}{2}k$.

Let us choose $\lambda = \frac{1}{2}$, then k must be $= 1$.

j \ i	0	1	2	3	4
0	0	3	4	3	0
1	0	2	3	2	0
2	0	1.5	2	1.5	0
3	0	1	1.5	1	0
4	0	75	1	.75	0
5	0	5	.75	.5	0

Range for x is 4, so we put $x = ih$, $x_i = ih = i$ $\quad (\because h = 1)$

and $\quad t = jh$, $t_j = jk = j$ $\quad (\because k = 1)$

Also, $\quad u(x, 0) = x(4 - x) = i(4 - i)$

Now for various values of $i(i = 0, 1, 2, 3, 4)$, at time $t = 0$, values of $u(x, 0)$ are 0, 3, 4, 3, 0 and are written in the first row of the table. Again $u(0, t) = 0$ for all values of t, so $u(0, j) = 0$ for all values of j. Thus the entries in the first column of the table are all zero. Similarly, $u(4, t) = 0$ for all values of t, *i.e.*, $u(4, j) = 0$ for all j, hence all the entries in the last column are again zero.

Further, by **Bender Schmidt's recurrence relation,** we have

$$u_{i,j+1} = \frac{1}{2}[u_{i+1,j} + u_{i-1,j}] \qquad \text{... (i)}$$

Now, putting $j = 0$ in (*i*), we readily obtain

$$u_{i,1} = \frac{1}{2}[u_{i+1,0} + u_{i-1,0}]$$

$$\Rightarrow \quad u_{1,1} = \frac{1}{2}(u_{2,0} + u_{0,0}) = \frac{1}{2}(4-0) = 2, \quad \text{... (2)}$$

$$u_{2,1} = \frac{1}{2}(u_{3,0} + u_{1,0}) = \frac{1}{2}(3+3) = 3,$$

and $$u_{3,1} = \frac{1}{2}(u_{4,0} + u_{2,0}) = \frac{1}{2}(0+4) = 2,$$

Thus the second row is complete.

Again putting $j = 1, 2, 3, 4$ the other rows can be easily completed.

11.9 DERIVE THE CRANK-NICHOLSON DIFFERENCE SCHEME FOR THE PARABOLIC EQUATION $u_{xx} = a\,u_t$ WITH BOUNDARY CONDITIONS AS $u(0, t) = T_0$, $u(l, t) = T_1$ AND THE INITIAL CONDITION AS $u(x, 0) = f(x)$

The given equation is $u_{xx} = au_t$

By 11.4 we have $$u_{xx} = \frac{u_{i+1,j+1} - 2u_{i,j+1} + u_{i-1,j+1} + u_{i+1,j} - 2u_{i,j} = u_{i-1,j}}{2h^2} \quad \text{... }(i)$$

and $$u_i = \frac{u_{i,j+1} - u_{i,j}}{k}$$

Then substituting in (i), we obtain

$$\frac{u_{i+1,j+1} - 2u_{i,j+1} + u_{i-1,j+1} + u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{2h^2} = \frac{a(u_{i,j} - u_{i,j})}{k}$$

$$\frac{k}{2h^2a}(u_{i+1,j+1} - 2u_{i,j+1} + u_{i-1,j+1} + u_{i+1,j} - 2u_{i,j} + u_{i-1,j}) = u_{i,j+1} - u_{i,j}$$

Putting $\frac{k}{h^2a} = r$, we have

$$\frac{1}{2}r\,u_{i+1,j+1} - r\,u_{i,j+1} + \frac{1}{2}r\,u_{i-1,j+1} - u_{i,j+1} = r\,u_{i,j} - \frac{1}{2}r\,u_{i+1,j} - \frac{1}{2}r\,u_{i-1,j} - u_{i,j}$$

$$\Rightarrow \frac{1}{2}r\,u_{i-1,j+1} - (r+1)\,u_{i,j+1} + \frac{1}{2}r\,u_{i-1,j+1} = -\frac{1}{2}r\,u_{i-1,j} + (r-1)\,u_{i,j} - \frac{1}{2}r\,u_{i+1,j}$$

$$\Rightarrow -r\,u_{i-1,j+1} + 2(r+1)\,u_{i,j+1} - r\,u_{i+1,j+1} = r\,u_{i-1,j} + 2(1-r)\,u_{i,j} + r\,u_{i+1,j} \quad \text{... }(ii)$$

This is called C.N. difference scheme.

11.10 HYPERBOLIC EQUATIONS

One dimensional wave equation is given by

$$a^2\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial t^2} = 0 \Rightarrow a^2 u_{xx} - u_{tt} = 0$$

where $$A = a^2, B = 0, C = -1.$$

$\therefore \quad B^2 - 4AC = 0 - 4a^2 \times -1 = 4a^2 + \text{+ve}$

$\therefore$ The equation is hyperbolic.

Let us solve, by the method of finite difference, the equation

$$a^2 u_{xx} - u_{tt} = 0 \qquad \text{... } (i)$$

under the boundary conditions:

$$u(x_0, t) = 0 \text{ ... } (i) \; ; \; u(l, t) = 0 \qquad \text{... } (ii)$$

and the initial conditions:

$$u(x, 0) = f(x) \qquad \text{... } (iii)$$

$$u_1(x, 0) = 0 \qquad \text{... } (iv)$$

Now, selecting the spacing h for the variable x and a spacing k for t ; we have

$$u_{xx} = \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2} \; ; \; u_{tt} = \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{h^2} \; ;$$

and $$u_t = \frac{u_{i,j+1} - u_{i,j}}{k}$$

Thus equation (1)

$$a^2 \frac{(u_{i+1,j} - 2u_{i,j} + u_{i-1,j})}{h^2} - \frac{(u_{i,j+1} - 2u_{i,j} - u_{i,j-1})}{k^2} = 0 \qquad \text{... } (i)$$

$$u_{i+1,j} - 2u_{i,j} + u_{i-1,j} - \frac{h^2}{k^2 a^2} (u_{i,j+1} - 2u_{i,j} + u_{i,j-1}) = 0$$

Now assuming $r = \left(\frac{k}{h}\right)$, we have

$$r^2 a^2 (u_{i+1,j} - 2u_{i,j} + u_{i-1,j}) - u_{i,j+1} + 2u_{i,j} - u_{i,j} - u_{i,j-1} = 0$$

$$\therefore \quad u_{i,j+1} = 2(1 - r^2 a^2)\, u_{i,j} + r^2 a^2 (u_{i+1,j} + u_{i-1,j}) - u_{i,j-1} \qquad \text{... } (ii)$$

Again, the boundary conditions (i) and (ii) can be put in a different form as

$$u_{0,j} = 0 = u_{n,j} \quad j = 1, 2, 3, (nh = l) \qquad \text{... } (iii)$$

Also, the initial condition (iii) can be as

$$u_{i,0} = f(ih) \; i = 1, 2, \ldots \qquad \text{... } (iv)$$

and (iv) can be written as

$$\frac{u_{i,j+1} - u_{i,j}}{k} = 0 \text{ when } t = 0, \text{ i.e., } j = 0$$

$$\Rightarrow \quad u_{i,1} = u_{i,0} = 0 \qquad \text{... } (v)$$

i.e., $$u_{i,1} = u_{i,0} = f(ih) \text{ using } (iv) \qquad \text{... } (vi)$$

Whence (iv) and (v) give the values of u on the first two rows $i = 0$ and $j = 1$.

Further, putting $j = i$ in (ii), we get

$$u_{i,2} = 2(1 - r^2 a^2)\, u_{i,1} + r^2 a^2 (u_{i+1,1} + u_{i-1,1}) - u_{i,0} \qquad \text{... } (vii)$$

Obviously, the terms on the R.H.S. of (vii) involve the values of u on the first two rows ($j = 0$ and $j = 1$) and these are known from the initial conditions (iv) and (vi), hence $u_{i,2}$ is calculated explicitly.

Then knowing $u_{i,2}$ we can calculate $u_{i,3}$ by putting $j = 2$ in (ii) and so on. Hence (ii) gives an explicit scheme for the solution to one dimensional wave equation.

If $k < h$, the solution (ii) is convergent and the coefficient of $u_{i,j}$ in (*ii*) will be zero if we take $1 - r^2 a^2 = 0$

$$\Rightarrow \qquad r^2 = \frac{1}{a^2} \Rightarrow \frac{k^2}{h^2} = \frac{1}{a} \Rightarrow k = \frac{h}{a}$$

Then, the recurrence relation (*ii*) takes the simplified form as:

$$u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1} \qquad \text{... (viii)}$$

Hence the value of u at $x = x_i$ at a time $t = t_j + k$ equals to the sum of the values of u at the surrounding points $x = x_{i-1}$ and $x = x_{i+1}$ at the previous time $t = t_j$ minus the value of at $x = x_i$ at time $t = t_j - k$.

SOLVED EXAMPLES

Example 11.13. Evaluate the pivotal values of the following equation taking $h = 1$ and upto one half of the period of vibration

$$16\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$$

given that $u(0, t) = u(5, t) = 0$; $u(x, 0) = x^2(5 - x)$ and $u_t(x, 0) = 0$.

Solution: $\dfrac{\partial^2 u}{\partial t^2} = a^2 \dfrac{\partial^2 u}{\partial x^2}$, is the equation of the vibrating string of length l, whose period of vibration $= \left(\dfrac{2l}{a}\right)$

Here $\qquad l = 5$ and $a^2 = 16$, *i.e.*, $a = 4$

$$\therefore \qquad \text{Period} = 2 \times \frac{5}{4} = \frac{5}{2} \text{ secs.}$$

Now, we want values of u upto $t = \dfrac{5}{4}$ seconds.

The given equation is $16u_{xx} = u_{tt}$ ... (*i*)

$$\Rightarrow \qquad \frac{16(u_{i+1,j} - 2u_{i,j} + u_{i-1,j})}{h^2} = \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{k^2} \qquad \text{(in terms of diff. expressions)}$$

Now, taking $h = 1$, we get

$$16k^2 (u_{i+1,j} - 2u_{i,j} + u_{i-1,j}) = u_{i,j+1} - 2u_{i,j} + u_{i,j-1}$$

$$\Rightarrow \qquad u_{i,j+1} = 2(1 - 16k^2)\, u_{i,j} + 16k^2(u_{i+1,j} + u_{i-1,j}) - u_{i,j-1}$$

If we choose k, so that the co-efficient of $u_{i,j} = 0$; then the above gives

$$1 - 16\,k^2 = 0 \Rightarrow k^2 = \frac{1}{16} \Rightarrow k = \frac{1}{4}$$

Then the above equation takes the form:

$$u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1} \qquad \text{... (ii)}$$

As $\qquad k = \dfrac{1}{4} < h = 1$, so the solution (*ii*) is convergent.

Also, the initial conditions $u(0, t) = 0 = u(5, t)$ become

$$\left.\begin{aligned} u_{0,j} &= 0 \\ \text{and } u_{n,j} &= 0 \end{aligned}\right\} \qquad \text{... (iii) for all values of } j.$$

Thus the first and last column entries giving the values of $u_{i,j}$ are all zero. The condition $u_t(x, 0) = 0$

$$\Rightarrow \quad \frac{u_{i,j+1} - u_{i,j}}{k} = 0 \text{ when } t = 0, \text{ i.e., when } j = 0$$

$$\Rightarrow \quad u_{i,1} - u_{i,0} = 0 \quad \text{or} \quad u_{i,1} = u_{i,0} \qquad \text{... (iii)}$$

Also the initial condition $u(x, 0) = x^2(5 - x)$ implies

$$u_{i,0} = i^2(5 - i) \qquad \text{...(iv)}$$

Then putting $i = 1, 2, 3, 4$ in (*iv*), we get

$$u_{1,0} = 1(5 - 1) = 4;\ u_{2,0} = 2^2(5 - 2) = 12;$$

$$u_{3,0} = 3^2(5 - 3) = 18 \text{ and } u_{4,0} = 4^2(5 - 4) = 16.$$

Thus the first row is now complete.

But $u_{i,1} = u_{i,0}$, so the second row is the same as the first row.

Again putting $j = 1$ in (*ii*) we get

$$u_{i,2} = u_{i+1,1} + u_{i-1,1} - u_{i,0} \qquad \text{... (v)}$$

Now putting $i = 1, 2, 3, 4$ successively in (*v*), we get

$$u_{1,2} = u_{2,1} + u_{0,1} - u_{1,2} = u_{2,0} + 0 - 4 = 12 - 4 = 8$$

$$u_{2,2} = u_{3,1} + u_{1,1} - u_{2,0} = u_{3,0} + u_{1,0} - 12 = 18 + 4 - 12 = 10$$

$$u_{3,2} = u_{4,1} + u_{2,1} - u_{3,0} = u_{4,0} + u_{2,0} - 18 = 16 + 12 - 18 = 10$$

$$u_{4,2} = u_{5,1} + u_{3,1} - u_{4,0} = u_{5,0} + u_{3,0} - u_{4,0} = 0 + 18 - 16 = 2$$

This completes the third row. In a similar manner, all the rows can be completed. The values of $u_{i,j}$ are given in the following table:

j \ i	0	1	2	3	4	5
0	0	4	12	18	16	0
1	0	4	12	18	16	0
2	0	0 + 12 – 4 = 8	4 + 18 – 12 = 10	12 + 16 – 18 = 10	18 + 0 – 16 = 2	0
3	0	0 + 10 – 4 = 6	8 + 10 – 12 = 6	10 + 2 – 18 = – 6	10 + 0 – 16 = – 6	0
4	0	– 2	– 10	– 10	– 8	0
5	0	– 16	– 18	– 12	– 4	0

EXERCISE 11.4

1. Solve $y_{tt} + y_{xx}$ upto $t = 0.5$ with a spacing of 0.1 subject to $y(0, t) = 0$, $y(1, t) = 0$, $y_t(x, 0)$ and $y(x, 0) = 10 + x(1 - x)$.

Ans.

j \ i	0	1	2	3	4	5	6	7	8	9	10
0	0	10.19	10.16	10.21	10.24	10.25	10.24	10.21	10.16	10.09	0
1	0	5.08	10.15	10.20	10.23	10.24	10.23	10.20	10.15	5.08	0
2	0	0.06	5.12	10.17	10.20	10.21	10.20	10.17	10.12	0.06	0
3	0	0.04	0.08	5.12	10.15	10.16	10.15	10.12	10.08	0.04	0
4	0	0.02	0.04	0.06	5.08	10.09	10.08	10.06	10.04	0.02	0
5	0	0	0	0	0	0	0	0	0	–0.02	0